

# Annex 3

## International Atomic Energy Agency and World Health Organization guidelines on good practices for quality control of in-house radiopharmaceutical preparations

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## Abbreviations

|       |                                                                                                     |
|-------|-----------------------------------------------------------------------------------------------------|
| ALARA | as low as reasonably achievable                                                                     |
| HPLC  | high-performance liquid chromatography                                                              |
| ICH   | International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use |
| TLC   | thin-layer chromatography                                                                           |
| UV    | ultraviolet                                                                                         |

## 1. Background

1.1 A robust quality control system tailored to the unique nature of radiopharmaceuticals requires careful consideration. Radiopharmaceuticals possess inherent characteristics that demand specialized attention, including the narrow testing time window, variability in the types of emitted ionizing radiation, complexities associated with the simultaneous production of radioactive drug substances and final drug products, and the constraints of radiation handling. These factors must be systematically integrated into the design of radiopharmaceutical quality control processes. Concurrently, however, radiopharmaceutical quality control testing must be sufficiently comprehensive and well integrated into the production process, as its function ensures that the radiopharmaceutical administered to the patient provides the intended benefit. These concepts must be incorporated into nuclear medicine professional training and practice. Along with these efforts, this guidance provides recommendations on the minimum requirements for establishing a radiopharmaceutical-specific quality control testing programme.

## 2. Scope

2.1 This guideline provides a general overview of good practices for the quality control of radiopharmaceuticals. This guideline is consistent with *WHO good practices for pharmaceutical quality control laboratories* ([1](#)) and the *International Atomic Energy Agency and World Health Organization guideline on good manufacturing practices for radiopharmaceutical products* ([2](#)). The recommendations in this guideline are applicable to the quality control of:

- starting materials;
- finished radiopharmaceutical products;
- radionuclides.

- 2.2 The requirements of this guideline apply to quality control testing of radiopharmaceutical products produced from starting materials under the practice of in-house production.
- 2.3 The requirements of this guidance do not cover quality control testing of cold kits prepared under the practice of in-house production. Those requirements are outlined in *IAEA/WHO good manufacturing practices for in house cold kits for radiopharmaceutical preparations* (3).
- 2.4 Compliance with the recommendations provided in these guidelines will help promote international harmonization of quality control laboratory practices for radiopharmaceuticals and will facilitate cooperation among laboratories and mutual recognition of results, as well as taking into account regulatory expectations. The good practice described herein serves as a general guide that may be adapted to meet individual needs, provided that an equivalent level of quality assurance is achieved.
- 2.5 This guidance primarily addresses the testing of the physicochemical properties of starting materials, finished radiopharmaceuticals products and radionuclides. Microbiology-related quality control testing falls outside its scope. For more information on microbiological quality control, refer to the *WHO good practices for pharmaceutical microbiology laboratories* (4).

### 3. Glossary

**acceptance criterion for an analytical result.** Predefined and documented indicator that determines whether a result falls within or outside the limits specified in the corresponding specification.

**accuracy.** The degree of agreement between the value obtained from an analytical procedure and a conventional true value or an accepted reference value.

**analytical test report.** A document that includes a description of the test procedures used, the results of the analysis, discussions and conclusions.

**as low as reasonably achievable (ALARA).** A principle of radiation protection aimed at minimizing radiation exposure to the lowest reasonable level. This is achieved through time management, maintaining distance, using shielding and promoting awareness among stakeholders.

**authenticated reference standard.** A well characterized material that can be used as a reference standard during calibration or quality control testing. The characterization of material must include confirmation or authentication of true value for the specification that will subsequently be used as reference. Authenticated material may or may not have accompanying certification.

**batch (non-radioactive material).** A defined quantity of material or product prepared in a single process or series of processes to ensure homogeneity.

**batch (radiopharmaceutical).** A quantity of a radiopharmaceutical product prepared in a single procedure or series of procedures to ensure homogeneity.

**calibration.** The set of operations performed under specified conditions to establish the relationship between measurements recorded by an instrument or system and the corresponding known values of a reference standard. Acceptable measurement limits should be defined.

**certificate of analysis.** A document listing the test procedures performed on a sample, the obtained results and the applied acceptance criteria, indicating whether the sample complies with the specification.

**end-of-synthesis or end-of-production time.** The time recorded when the very last step in radiopharmaceutical batch production (for example, sterile filtration or formulation) is completed.

**good manufacturing practices for radiopharmaceuticals.** A set of guidelines ensuring that radiopharmaceutical products are consistently produced and controlled according to established quality standards. Good manufacturing practice is part of an overall quality management system to ensure product traceability and consistency.

**installation qualification.** Testing performed to verify that analytical equipment is installed correctly and operates according to established specifications.

**operational qualification.** A documented verification process to confirm that analytical equipment functions as intended across all anticipated operating ranges.

**out-of-specification result.** A test result that falls outside the established specifications or acceptance criteria.

**performance qualification.** A documented process to verify that analytical equipment operates consistently and produces reproducible results within defined specifications and parameters for its intended analytical application.

**precision.** The degree of agreement among individual test results when the procedure is applied repeatedly to multiple samplings of a homogeneous sample. Precision is typically expressed as relative standard deviation and can be considered at three levels: repeatability (precision under the same operating conditions over a short period of time), intermediate precision (within laboratory variations – different days, different analysts or different equipment), and reproducibility (precision between laboratories).

**qualified supplier.** Established and reputable firm responsible for the production and supply of standardized starting materials (for example, reagents, reference standards, automated system cassettes, sterilized vials and purification cartridges) commonly used in radiopharmaceutical preparation for clinical use that have been successfully evaluated by the radiopharmaceutical producer, confirming presence of an adequate quality system.

**quality control.** A set of analytical tests performed to verify compliance with predetermined quality acceptance criteria for starting materials, radionuclides and finished radiopharmaceutical products.

**quality management system.** A structured system that includes an organization's procedures, processes and resources to ensure that a radiopharmaceutical product or service meets specified quality requirements.

**radiopharmaceutical product.** Any pharmaceutical product that, when ready for use, contains one or more radionuclides (radioactive isotopes) for medicinal purposes.

**reference substance (or standard).** An authenticated material of uniform composition used in chemical and physical tests, allowing comparison with the product under examination. It must have an appropriate level of purity and accurate assigned value for its intended use.

**specifications.** A list of detailed requirements (acceptance criteria for the prescribed test procedures) with which the starting material, the radionuclide or the radiopharmaceutical product must conform in order to ensure suitable quality.

**standard operating procedure.** A written, authorized document providing detailed instructions for performing general or specific operations.

**system suitability test.** A test conducted on an analytical instrument using a reference standard or standards prior to radiopharmaceutical product quality control sample analysis to verify that the analytical instrument functions as intended for the intended analysis.

**validation of an analytical procedure.** A documented process by which a non-pharmacopoeial analytical procedure (or method) is demonstrated to be suitable for its intended use.

## 4. Quality management system

4.1 A quality management system should be in place to ensure the proper control of radiopharmaceutical products. It should encompass, among

other elements, the organizational structure of the quality control laboratory, including job descriptions, procedures, processes, resources and actions necessary to provide adequate confidence that radiopharmaceutical products are properly controlled and released only if they meet the intended quality standards.

- 4.2 Risk management principles should be integrated in the establishment, implementation and management of the quality management system.
- 4.3 Radiopharmaceuticals constitute a specific class of pharmaceuticals with unique characteristics that should be considered when implementing a quality management system:
  - simple distribution chain, whereby the finished product is often delivered directly from the manufacturer to the nuclear medicine department, which may also serve as the manufacturer;
  - small batch size;
  - quality control sample representing the entire batch of radiopharmaceutical product;
  - limited shelf-life, ranging from minutes to several days;
  - decay characteristics of the intended radionuclides.
- 4.4 Radiopharmaceuticals with limited shelf-life are often administered prior to completion of all quality control testing. Tests such as sterility, endotoxin content and radionuclidic purity determination may need to be performed post-release. Hence, the importance of good practices and a robust quality management system are essential to minimize the possible risks to quality that may not be identified through quality control pre-release testing.
- 4.5 Quality control of radiopharmaceuticals, particularly those labelled with short half-life radionuclides, is typically performed soon after preparation. Since quality control laboratories are often integrated within the same manufacturing infrastructure, quality control activities generally fall under the broader scope of the overall quality management system.
- 4.6 In alignment with general quality management system principles, documentation summarizing the following information should be established:
  - the organizational structure;
  - the personnel involved and their respective responsibilities and duties, including training;

- a description of the laboratory;
- a list of the available equipment;
- the qualification policy for relevant instruments;
- the validation and verification policy for analytical methods, if applicable.

- 4.7 If quality control laboratories operate independently from radiopharmaceutical manufacturers (for example, in countries where regulatory agencies maintain dedicated quality control laboratories for verifying the quality of candidate drugs for marketing authorization), a specific quality management system should be implemented. This system should adhere to the same principles outlined above to ensure consistency and reliability.
- 4.8 The quality control of radiopharmaceuticals must comply with applicable national radiation safety regulations and be conducted in accordance with the as low as reasonably achievable (ALARA) principles ([5](#), [6](#)).

## 5. Personnel and training

- 5.1 The quality control laboratory should have a sufficient number of qualified personnel to effectively carry out its intended operations.
- 5.2 The laboratory should maintain clearly defined job descriptions for all personnel involved in quality control testing, as well as in the calibration, validation and qualification of equipment, and verification of analytical procedures.
- 5.3 The quality control laboratory and its personnel should operate under the supervision of a responsible person with appropriate qualifications and experience, as required by national regulations.
- 5.4 The responsible person should oversee batch release and the approval of certificates of analysis to ensure compliance with established quality standards.
- 5.5 Personnel must possess appropriate qualifications, training and experience relevant to their specific responsibilities and job descriptions, as determined by the responsible person.
- 5.6 A documented training programme should be in place. Training should include the handling of radioactive materials and radiation safety protocols. Personnel should undergo regular training and periodic courses to stay updated on the latest advancements in their field.



- 5.7 All training activities should be documented, and records should be maintained for future reference.
- 5.8 All personnel handling radioactive materials must undergo regular monitoring for potential contamination and radiation exposure in accordance with radiation safety regulations.

## 6. Premises

- 6.1 Quality control laboratories should be strategically located and properly designed, constructed, adapted and maintained to accommodate the specific operations to be performed.
- 6.2 Quality control laboratories should be appropriately segregated from preparation laboratories to prevent cross-contamination and ensure independent testing conditions.
- 6.3 A dedicated area for handling and storage of radioactive samples should be specially designed, properly equipped, and adequately shielded and ventilated based on the decay characteristics of the radionuclides used. Radiation protection measures should be implemented in accordance with ALARA principles to ensure the safety of operators.
- 6.4 Equipment, or relevant components thereof, used for radioanalytical quality control (for example, radio-high-performance liquid chromatography, radio-thin-layer chromatography, gamma spectrometer and dose calibrator) should be properly shielded to maintain low background signal levels for sensitive radioactive detectors and minimize radiation exposure to operators.
- 6.5 The heating, ventilation and air-conditioning system should be designed to maintain appropriate temperature and ventilation levels. Unlike manufacturing areas, quality control laboratories typically do not require classification according to good manufacturing practice rules and grades.
- 6.6 Radioactive gas emissions should be effectively controlled and monitored in order to minimize the risk of unnecessary radiation exposure to personnel and the surrounding environment. Radiation detectors with alarm systems should be installed to detect potential leaks. Dedicated fume hoods with negative pressure maintenance should be used where necessary (for example, near gas chromatography system exhaust line outlets, where potentially contaminated vapours may be released during analysis).
- 6.7 Special precautions should be taken when handling highly toxic or radiotoxic substances. If required, a separate dedicated unit or containment equipment

(for example, isolators) should be used to prevent exposure and contamination. Proper procedures should be in place to ensure operator safety.

- 6.8 Drains and sinks are allowed in quality control laboratories, provided that they are appropriately designed.
- 6.9 Quality control laboratories should either be equipped with or be in reasonably close proximity to shower facilities that could be used as emergency showers in case of emergencies.
- 6.10 Suitable shielded containers should be in place to dispose of contaminated materials, such as needles and syringes. The waste containers should be separated according to the decay properties of the radionuclide (alpha, beta, etc.) and half-life, if necessary.
- 6.11 Access to quality control work areas should be restricted to authorized personnel.

## 7. Equipment

- 7.1 The quality control laboratory should have the required apparatus, equipment, instruments or instrument systems for the correct performance of the tests and related activities (Z).
- 7.2 All equipment maintenance, qualification and calibration activities should be documented, and records should be maintained for traceability and compliance purposes.
- 7.3 Computerized systems used for data acquisition and equipment control should be considered an integral part of the equipment or an instrument.
- 7.4 The dose calibrator should be calibrated and periodically checked using suitable reference standards. If a nationally recognized reference standard is unavailable, alternative sources – such as dose calibrator manufacturer model-specific recommendations that have been verified using the radionuclide sources supplied by radionuclide suppliers or published literature – may be considered when determining the appropriate dial setting.
- 7.5 Quality control laboratories may choose to rely on either electronic or paper-based quality management systems. In paper-based quality management systems, the standard operating procedure documents, batch records, and quality control instrument-generated data are stored in paper document form. The analytical instrument software may or may not be equipped with data integrity and audit trail features, as they may not always be available

for some of the instruments (for example, gamma counters or gamma spectrometers). The traceability and data integrity of all data generated by analytical instruments in a paper-based system are assured through implementation of an adequate system of good documentation practices.

- 7.6 In quality control laboratories that choose to rely on an electronic-based quality management system, or computer systems connected to quality control equipment for data collection, processing, recording, reporting, storage or retrieval, the following measures should be implemented (8).
- Procedures should be established to ensure data integrity and prevent unauthorized access. Whenever possible, software should include an audit trail feature to track changes.
  - User access to both the operating system and acquisition or control software should be managed through appropriate credentials based on user roles, preventing unauthorized modifications (for example, changes to methods, records or system clock).
  - Electronic data should be backed up at appropriate regular intervals according to a documented procedure. Backed-up data should be retrievable and stored in such a manner as to prevent data loss.
- 7.7 In quality control laboratories that choose to rely on a paper-based quality management system, good written documentation practices and standard operating procedures must be established and implemented to ensure integrity and traceability of the generated quality control data.

## 8. Qualification and calibration

- 8.1 The designated responsible person in a quality control laboratory should be responsible for selecting the analytical instruments and equipment, and their features (for example, type of detector), that are appropriate for the intended use.
- 8.2 Instruments for quality control should be qualified according to a defined qualification plan, which includes installation qualification, operational qualification and performance qualification (7).
- 8.3 The qualification process should demonstrate that quality control instruments are installed, are operated and perform as intended, and are suitable for their intended use.
- 8.4 A regular calibration schedule should be established. The frequency and methodology of calibration must be documented in written standard operating procedures.

- 8.5 All equipment, instruments and devices requiring calibration should be clearly labelled, coded or otherwise identified to indicate calibration status, date of last calibration and next scheduled recalibration date.
- 8.6 Measuring equipment (for example, calibrated thermometers or flow meters) should be calibrated regularly according to a predefined calibration plan established by the laboratory.
- 8.7 Specific procedures and frequencies of calibration should be established for each type of equipment. For example:
- pH meters should be verified with certified standard buffer solutions before use.
  - Balances should be checked prior to each use using certified reference weights. In balances equipped with an internal adjustment feature, recommendations from the weight balance manufacturer should be followed in regard to weight balance calibration and suitability checks.
  - Dose calibrators should be verified daily and calibrated periodically using certified calibration sources.
  - Radiation detectors (for example, high-performance liquid chromatography (HPLC), thin-layer chromatography (TLC) or gamma spectrometer) should be calibrated periodically and checked for suitability prior to analysis, to ensure they are fit for the intended analysis.
  - Other types of detectors (for example, ultraviolet (UV) detectors or flame ionization detectors) should be calibrated periodically using reference sources (for example, reference substances with known absorbance for UV cells).
- 8.8 Qualification and calibration documentation should include at least the following:
- identification details of the equipment, instrument or device;
  - manufacturer's information, including model, serial number or other unique identifiers;
  - required qualification, verification and calibration procedures;
  - dates, results, and copies of reports, verifications and calibration certificates;
  - acceptance criteria and next scheduled qualification, verification or calibration;
  - maintenance history and the future maintenance plan.

- 8.9 Analysis results should be traceable, where applicable, to a primary reference substance to ensure accuracy and reliability.
- 8.10 All calibration and qualification processes should be traceable to certified or authenticated reference materials and linked to SI units (metrological traceability) to maintain measurement accuracy.

## 9. Documentation

- 9.1 Good documentation practices should be followed for all quality control activities (8).
- 9.2 Documentation should ensure full traceability of radiopharmaceutical quality control processes.
- 9.3 Quality control documentation is an integral part of the batch record, providing a clear and complete account of the drug product characterization.
- 9.4 A controlled system of standard operating procedures should be established to cover all operations performed in a quality control laboratory, including:
- use and maintenance of quality control equipment;
  - qualification and calibration of quality control equipment;
  - management, backup and archiving of data;
  - description of analytical methods used;
  - handling of out-of-specification results;
  - validation policy for non-pharmacopoeial or non-compendial analytical methods;
  - quality control testing procedures for specific products or raw materials;
  - the use and storage of suitable reference substances and materials;
  - batch release records;
  - supplier qualification.
- 9.5 Maintenance and calibration plans should be defined periodically (for example, at the beginning of the year), detailing frequency and type of interventions required.
- 9.6 The standard operating procedures should be approved, signed and dated by the appropriate responsible personnel, and they should not be modified without proper review, evaluation and approval by the designated responsible individual. The standard operating procedures should be reviewed periodically to ensure their continued applicability and accuracy.

- 9.7 All documentation should be retained for an appropriate period based on its content, typically no less than two years or as required by local regulations.
- 9.8 Documentation is an essential component of the quality management system. The laboratory should establish and maintain procedures to control, review and update all quality-related documentation.
- 9.9 The procedures to control and review documents should ensure that:
- revised documents are referenced to previous versions to maintain traceability;
  - old or invalid documents are archived to document procedural evolution, while hard copies of obsolete documents are securely destroyed;
  - all relevant staff receive training on new and revised standard operating procedures.
- 9.10 All relevant quality control results for each batch of radiopharmaceuticals should be summarized in approved quality control documentation (for example, certificate of analysis, quality control batch record or quality control report). The documentation should contain, at a minimum:
- name and address of the quality control laboratory;
  - approved document identification number;
  - name and pharmaceutical form of the radiopharmaceutical;
  - batch number;
  - the name of the manufacturer or provider of the sample, if a quality control sample is provided by an external entity;
  - the radioactive concentration or batch radioactivity and volume at end-of-synthesis or end-of-production time, either at completion of the quality control analysis or if the quality control is to determine the amount of radioactivity (for example, using gamma spectrometry to determine the amount of radioactivity in <sup>225</sup>Ac-labelled products);
  - batch expiration date and time, if determined by quality control testing results;
  - results of all tests performed, including a comparison with established acceptance criteria (limits);
  - for each test, a conclusion indicating whether the result meets the acceptance criteria;
  - relevant comments, observations or information regarding specific test conditions necessary for result interpretation;

- final conclusion on whether the sample meets specifications and may be released or rejected, if applicable;
- date and signature of the responsible person.

## **10. Quality control of starting materials during material acceptance process**

- 10.1 The material acceptance process – including physical receipt, visual inspection, quality control testing and retesting, segregation requirements, storage conditions, expiration assignment and labelling – should be conducted in accordance with written and implemented standard operating procedures.
- 10.2 For materials supplied by qualified suppliers, acceptance may be based on the review and verification of the certificate of analysis to ensure conformance with internally established acceptance criteria. No additional testing applied by the quality control laboratory is required, as long as the results of a documented risk assessment support not performing the additional physical testing.
- 10.3 If starting materials are prepared in-house, testing criteria and analytical methods must be established and documented to accurately determine their identity and chemical purity.
- 10.4 For radionuclides supplied by a qualified supplier, acceptance may be based solely on the review of the associated certificate of analysis, and no further testing is required. The decision to not conduct any additional testing should be supported by a documented risk assessment.
- 10.5 For radionuclide generators supplied by a qualified supplier, acceptance may be based on a review of the related certificate of analysis, with additional testing performed as per the manufacturer's recommendations.
- 10.6 For radionuclides that are used in continuous processes and not isolated, quality assessment should be performed during production process validation. This assessment should follow internal specifications or conform to a recognized pharmacopoeia monograph.

## **11. Quality control of finished radiopharmaceutical products**

- 11.1 Finished radiopharmaceutical products are radioactive products that have undergone all stages of preparation, purification, formulation, quality control testing and packaging in their final container.

- 11.2 Each batch of a radiopharmaceutical product must undergo testing, as specified in the relevant written quality control standard operating procedures.
- 11.3 Due to the inherent radioactive properties of these products, certain quality control tests may be performed post-release. This is typically required for radionuclidic purity testing, which usually needs to be conducted after the complete decay of the desired radionuclide. This ensures that gamma spectrometry analysis can accurately detect and quantify any potential contaminant radionuclides. Under certain circumstances (for example, radiopharmaceutical products containing radionuclides with a short half-life or with high activity), post-release endotoxin content determination may be justified. The decision to perform this test post-release should be based on a risk assessment, unless specified in a pharmacopoeial monograph.
- 11.4 For radiation safety reasons, final product pH testing may be conducted using pH paper or strips, unless the pharmacopoeial monograph has a specific requirement for using a pH meter.
- 11.5 Quality control testing of radiopharmaceuticals containing radionuclides emitting low energy gamma, mixed energy gamma, beta, and alpha radiation (for example,  $^{124}\text{I}$ ,  $^{177}\text{Lu}$ ,  $^{161}\text{Tb}$ ,  $^{212}\text{Pb}$ ,  $^{225}\text{Ac}$ ) may require additional consideration and controls. At a minimum:
  - When measuring radioactivity in a dose calibrator, a dial setting specific to the geometry (that is, a specific type of vial that contains either the final product or quality control sample) that is used to measure the radioactivity at a specific facility should be established. This can be accomplished either through verification of reference sources, if available, or through conduct of high-purity germanium detector gamma spectrometry studies and subsequent cross-calibration.
  - When measuring radioactivity of radiopharmaceuticals containing radionuclides with radioactive progeny (for example,  $^{225}\text{Ac}$  or  $^{212}\text{Pb}$ ) in a dose calibrator, the measurements may need to be conducted once the radioactive parent–daughter equilibrium is reached.
  - HPLC testing of radiopharmaceuticals containing non-gamma-emitting parent radionuclides with gamma-emitting radioactive progeny (for example,  $^{225}\text{Ac}$ ) requires fraction collection and counting using a calibrated gamma counter, as the direct accurate detection of alpha radiation by the radioactivity detectors currently available is not possible. The frequency of fraction collection should



be clearly defined in written standard operating procedures. In addition, the HPLC system used for quality control of alpha emitters must be completely segregated and only used for  $^{225}\text{Ac}$ -labelled products, to avoid cross-contamination with radionuclides with higher energies that will interfere with accurate measurements.

- An alternative method to HPLC for  $^{225}\text{Ac}$ -labelled products is TLC, which offers the possibility of direct detection and measurement of  $^{225}\text{Ac}$ -labelled samples using the newly developed alpha probe-equipped TLC scanners (see ZnS(Tl) scintillator coupled PMT (photomultiplier tube) probes), in contrast to methods that require daughter element equilibration. In this case, care must be taken that the detector physically measures the thin film strip in sufficient proximity. This immediate measurement capability may be an important factor in the quality control design of  $^{225}\text{Ac}$ -labelled radiopharmaceuticals.
- High-purity germanium detector gamma spectrometers used to measure the radioactivity must be appropriately calibrated for energy, peak shape and efficiency using an accurate multisource reference standard. Suitability testing must be conducted on the gamma spectrometer prior to each analysis. The geometry of the efficiency calibration reference standard must be identical to the geometry of the quality control sample.

## 12. Reagents and standards

### 12.1 Reagents

12.1 All reagents and chemicals, including solvents and materials used in tests and assays, should be of an appropriate quality and suitable for the intended use ([1](#)).

12.2 Reagents should be purchased from qualified suppliers and should be accompanied by the certificate of analysis and the material safety data sheet, if required.

12.3 For the preparation of reagent solutions in the laboratory:

- responsibility for reagent preparation should be clearly defined in the job description of the assigned personnel;
- prescribed procedures should be followed, adhering to pharmacopoeial standards or other recognized guidelines, where available;

- records should be maintained for the preparation and standardization of volumetric solutions.

12.4 All reagent labels should clearly indicate:

- contents (name and composition);
- manufacturer's details;
- date of receipt and date of container opening;
- concentration or radioactivity measurement details, if applicable;
- storage conditions and, if applicable, any specific protection measures (such as protection from heat, light or atmosphere);
- expiry date or retest date.

12.5 For in-house prepared solutions, labels should specify:

- name of the solution;
- date of preparation and initials of technician or analyst;
- molarity (or concentration), if applicable;
- expiry date or retest date, as justified;
- concentration or radioactivity measurement details, if applicable.

12.6 Reagents must be stored under appropriate conditions as defined by the manufacturer, including temperature and humidity control. For radioactive solutions, appropriate shielding must also be ensured.

## 12.2 Reference standards

12.7 Reference standards may be used for testing samples of finished radiopharmaceutical products. A typical radiopharmaceutical reference standard is the non-radioactive ("cold") reference compound of the radiopharmaceutical (for example, [ $^{19}\text{F}$ ]FDG for [ $^{18}\text{F}$ ]FDG), which may be used for identification or quantification purposes (such as constructing a calibration curve). If available, reference standards for impurities might also be considered. Stability of reference standards in solution needs to be established, unless provided by the supplier of the standards.

12.8 For radiopharmaceuticals labelled with elements that lack naturally occurring non-radioactive isotopes (for example, technetium, astatine or actinium), an appropriate alternative reference standard may be used, such as:

- the precursor compound;
- a compound labelled with another element that results in a final compound with similar physicochemical properties.

The choice of reference standard should be scientifically justified on a case-by-case basis.

- 12.9 Reference substances are also used for the calibration of radioactivity detectors, gamma spectrometer detectors and dose calibrators. These typically consist of calibrated sources containing long-lived radionuclides that emit suitable energies. Such sources may include:

- a single radionuclide source (for example, caesium-137) for routine dose calibrator checks;
- a multi-nuclide source, containing several radionuclides for high-purity germanium detector energy and efficiency calibration;
- a multi-gamma source, which is a single radionuclide emitting across a broad spectrum of energies, commonly used for gamma spectrometer energy calibration.

The selection of calibration sources should be made case by case, depending on the intended analysis and radionuclides involved.

- 12.10 Other useful reference standards, as previously mentioned, include:

- buffer solutions for pH meter calibration;
- certified reference weights for analytical balance calibration;
- substances with known absorbance for the calibration of UV detectors.

- 12.11 Each reference substance and radioactive calibrated source should be traceable via a unique identification number or reference.

- 12.12 The identification number or reference must be documented in the analytical test report each time the reference substance is used.

## 13. Testing and methodology

- 13.1 The analytical tests to be performed should be determined based on a thorough evaluation of various factors, including:

- the type of sample being analysed (for example, a finished radiopharmaceutical product, radionuclide or starting material);
- the respective specifications and quality standards;
- the radioactive half-life;
- the type of radiation being emitted;
- other relevant physicochemical properties.

Pharmacopoeia monographs may provide an essential reference for selecting certain required tests. Not all tests described in monographs may be mandatory, dependent on specific product characteristics (such as production route or composition of formulation).

- 13.2 Basic physicochemical quality control tests for radiopharmaceuticals may be classified as follows.

### Tests for identification

The identity of a radiopharmaceutical cannot be verified using conventional direct analytical methods, such as nuclear magnetic resonance or mass spectrometry.

As a result, radiopharmaceutical identification is typically performed using indirect testing methods that either:

- compare the radiopharmaceutical with a suitable reference standard; or
- utilize the decay properties of the radionuclides for verification.

The identification tests aim to confirm both:

- the identity of the radionuclide (the “radio” component); and
- the identity of the molecular structure binding the radionuclide (the “pharmaceutical” component).

### Key identification tests

- Half-life determination:
  - Typically performed using dose calibrators, which verify that the time/activity curve of the sample matches the known half-life of the radionuclide.
  - Gamma spectrometers may also be used for half-life determination, although they are primarily employed for energy emission analysis (see below).
- Energy emission spectrum analysis (gamma spectrometry):
  - Gamma spectrometers determine the emitted energy spectrum, which provides a unique “fingerprint” for each radionuclide. The test that is required for identity testing of the radionuclide can be found in respective monographs.

- **Molecular structure identification:**
  - Typically verified using HPLC equipped with a dual detector system (a radioactivity detector and a detector based on mass determination, such as UV or electrochemical or refractive index detectors).
  - The retention time of the primary peak in the radiochromatogram is compared with the retention time of the UV peak obtained from an authentic non-radioactive reference standard.

### **Alternative identification methods**

Other techniques may be used depending on the nature of the radiopharmaceutical, such as:

- TLC;
- colorimetric assay;
- electrophoresis;
- particle size determination;
- radioimmunoassay fraction determination assay, combined with size exclusion high-performance liquid chromatography.

### **Impurity identification and quantification methods**

Tests for impurities aim to quantify the percentage or amount of impurities present in the final radiopharmaceutical preparation. Impurities in radiopharmaceuticals can be classified into the following categories.

#### **Radionuclidic impurities**

- These impurities refer to radioactive contaminants that originate from radionuclides other than the intended radionuclide.
- **Detection method:**
  - Often determined using gamma spectrometry, which identifies emitted energy spectra characteristic of contaminant radionuclides, once the desired radionuclide has decayed.
  - Dose calibrators may also be used in specific cases, such as in the determination of molybdenum-99 breakthrough in the routine control of technetium-99m generators.

## Radiochemical impurities

- These impurities refer to chemical forms of the intended radionuclide that are different from the desired radiopharmaceutical compound. Their presence affects radiochemical purity, which is crucial for ensuring proper biodistribution and targeting of the radiopharmaceutical.
- Detection method:
  - Typically analysed using HPLC equipped with a radiation detector or radio-thin-layer chromatography.
  - Alternative methods may be used depending on the specific radiopharmaceutical.
  - Filtration tests may be carried out for human albumin macroaggregates, ensuring the proper particle size distribution.

## Chemical impurities

- The nature of non-radioactive impurities varies depending on the specific radiopharmaceutical synthesis route and may include:
  - the non-reacted substrates and products of the decomposition;
  - by-products generated during the radiosynthetic process;
  - metal catalysts;
  - metallic impurities;
  - phase transfer catalysts (for example, Kryptofix® in fluorine-18 chemistry);
  - residual solvents.
- Detection method:
  - Analytical methods for chemical impurities vary widely and depend on the nature of the impurity and the radiopharmaceutical formulation. Examples are residual solvents determined by gas chromatography with flame ionization detection or phase-transfer catalysis, as determined using colorimetric methods.
  - As a result, a detailed discussion of chemical impurity testing methods falls outside the scope of this guideline.

## 14. Sampling

- 14.1 Sampling should be representative of the batch from which the samples are taken. It must be conducted in a manner that prevents contamination, mix-ups or any other adverse effects on quality (9).
- 14.2 Sampling procedures should be adapted based on the type of material being analysed. For non-radioactive starting materials, standard sampling procedures and criteria may be followed.
- 14.3 In the majority of situations, radiopharmaceuticals are present as clear and homogeneous solutions packaged in glass vials. Sampling generally involves withdrawing an aliquot of the solution in a volume sufficient for the intended analysis. In such a scenario, an aliquot is representative of the entire batch and random sampling is not required.
- 14.4 The container holding radioactive samples should be stored with adequate radiation shielding and should be prelabelled with essential information, including at a minimum:
  - date and time of sampling (if applicable);
  - name of the radiopharmaceutical or radionuclide;
  - batch number.
- 14.5 Sampling of radioactive material should consider ALARA principles. Whenever possible, automated sampling systems located in suitably shielded environments (for example, automated dispensing systems) should be used.
- 14.6 In cases where batch size is limited (often a single vial), sampling may need to be performed manually by withdrawing an aliquot using a syringe. In such cases, the batch vial should be placed behind a suitable shielding barrier to protect the operator from radiation exposure.
- 14.7 Samples should be of an adequate volume to allow for the completion of all necessary quality control tests. A single sample may be withdrawn from the finished radiopharmaceutical vial and then distributed into multiple aliquots, each designated for a specific analytical test.
- 14.8 For radiopharmaceuticals in gaseous form, sampling should be performed using a suitable collection system that ensures proper delivery of the sample to the analytical instrument (for example, a gas chromatograph).
- 14.9 If necessary, a sample may be diluted with an appropriate diluent to achieve suitable condition for analysis.

- 14.10 Quality control samples do not need to be retained unless there is a viable and scientifically justified method and a need to retest samples in the future. In practice, because most radiopharmaceutical products are very short lived, due to either short radioactive half-life or continuous autoradiolysis, quality control sample retention for the majority of radiopharmaceutical products is not required.

## 15. Validation of analytical procedures

- 15.1 All analytical procedures used for quality control testing of radiopharmaceutical products must be suitable for their intended purpose. The analytical methods applied to quality control of radiopharmaceuticals may be divided into two general categories: (a) analytical methods described in relevant pharmacopoeia chapters or monographs, either as a stand-alone method or as part of quality control testing of a specific product; or (b) analytical methods not described in any pharmacopoeia.
- 15.2 In general, compendial or pharmacopoeial analytical methods described in regionally recognized pharmacopoeia chapters do not require additional validation. In the case of complex analytical procedures (such as HPLC testing), there may be the need to verify that a general compendial method may be suitable for a specific product or material being tested.
- 15.3 Non-compendial analytical methods applied to quality control testing of radiopharmaceuticals prepared in-house should require method validation to ensure those methods are suitable for their intended purpose. Validation also establishes acceptance criteria for system suitability tests, which are subsequently performed to verify the analytical procedure before analysis.
- 15.4 Validation should be conducted according to an approved validation protocol, which outlines the analytical performance characteristics to be evaluated based on the type of analytical procedure.
- For validation of non-radioactive products, validation parameters should align with recognized standards or guidelines (for example, WHO recommendations, as well as recommendations of the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH) guidelines on analytical method validation) ([1,10, 11](#)).
  - For radioactive products, validation parameters should align with regionally recognized standards or guidelines (for example,



guidelines of the European Association of Nuclear Medicine on the validation of analytical methods for radiopharmaceuticals) ([12](#), [13](#)). It should be recognized, however, that the published guidelines are only recommendations and should be applied whenever scientifically possible. The ability to establish analytical method validation parameters is ultimately dependent on the specific radioactive material being tested and the type of analytical method being applied. Therefore, the relevant analytical method validation parameters may need to be decided on a case-by-case basis, if compliance with the published recommendations is not possible.

All validation results must be documented in a manner that ensures traceability.

- 15.5 Analytical method revalidation is required whenever any changes that render the method no longer fit for the intended purpose occur. These changes may be related to the material being tested (for example, changes in sample analyte concentration or chemical composition) or in the analytical method itself (for example, changes in analytical instruments being used or changes in methods of analysis). The need for revalidation should be decided by a responsible person based on risk assessment on a case-by-case basis.
- 15.6 System suitability testing is an essential component of many analytical procedures. The specific system suitability tests required depend on the type of analytical method used.
  - These tests should be conducted before quality control sample analysis using appropriate reference standards.
  - If the system suitability criteria are met, the instrument is considered fit for its intended analysis.

## 16. Specifications

- 16.1 For substances or radiopharmaceuticals with an official monograph in *The International Pharmacopoeia*, the specifications may be defined based on the respective pharmacopoeial monograph.
- 16.2 For substances or radiopharmaceuticals that are covered by other recognized pharmacopoeias (for example, European Pharmacopoeia or United States Pharmacopeia) or by publicly available assessed dossiers (for example, for clinical trials or marketing authorization applications), these references may be adopted after a thorough evaluation of the proposed specifications to ensure suitability.

16.3 For substances or radiopharmaceuticals without an existing pharmacopoeial reference (for example, newly developed radiopharmaceuticals or active pharmaceutical ingredients), specifications should be established based on:

- comprehensive knowledge of the substance or radiopharmaceutical;
- validation data collected during radiopharmaceutical development;
- reference to general monographs, such as *The International Pharmacopoeia* general monograph on radiopharmaceuticals ([14](#)), where applicable.

16.4 For finished radiopharmaceutical products, the following testing parameters or characteristics should be considered:

- appearance;
- pH;
- identification and quantification of the non-radioactive (cold) counterpart (for example, via HPLC or another suitable technique), if applicable;
- identification and quantification of impurities, if applicable;
- identification and quantification of excipients, if applicable;
- identification of the radionuclide (for example, via gamma spectrometry, including approximate half-life determination);
- excipients (for example, guidelines for specific excipients such as ethanol should be followed) ([15](#));
- quantification of unknown peaks seen on the HPLC UV trace (“chemical purity”), if applicable;
- radiochemical identity;
- radiochemical purity;
- radionuclidic purity: this test does not need to be conducted on every batch or radiopharmaceutical product and may be done periodically, if warranted (for example, for in-house cyclotron produced or in-house radiochemically isolated radionuclides). For radionuclides supplied from qualified suppliers, the radionuclidic purity data provided by the radionuclide supplier may be used by the producer, with no additional testing required. For generator-produced radionuclides, the radio nuclidic purity testing should follow the recommendations from the generator supplier;
- specific activity at end-of-synthesis or end-of-production time, if applicable;
- endotoxin content;

- sterilizing filter integrity testing;
  - radioactive concentration at end-of-synthesis or end-of-production time;
  - dissolution time, if applicable (for example, for solid radiopharmaceuticals such as  $^{131}\text{I}$  labelled capsules);
  - radioimmunoactivity, if applicable (for example, radiolabelled monoclonal antibodies).
- 16.5 Specifications and acceptance criteria must be met throughout the entire shelf-life of the radiopharmaceutical.
- 16.6 Out-of-specification results must be thoroughly investigated by the responsible person in the quality control laboratory, with support from analysts. The outcome of the investigation should be documented and archived.
- 16.7 A dedicated standard operating procedure for investigation of out-of-specification results must be in place. The standard operating procedures should outline clear steps, typically including:
- verification of the operations executed by the operators;
  - review of specifications and analytical methods;
  - check of calculations, if applicable;
  - re-evaluation of all results;
  - examination of reference standards, if used;
  - verification of instrument performance;
  - review of instrument calibration status;
  - evaluation of system suitability test results.
- 16.8 If the out-of-specification result is determined to be caused by a clear and identifiable error during the initial analysis it may be invalidated, and the testing may be repeated. The batch may still be released, based on retesting results. The decision to retest and to subsequently release the batch should be made by the responsible person.
- 16.9 If an out-of-specification result is confirmed, the batch of the starting material, radionuclide or radiopharmaceutical must be rejected or recalled if it has been supplied to a user.
- 16.10 In cases where results are inconclusive, resampling or retesting is permitted if there is clear evidence that the initial results were invalid. This decision should be made by the responsible person.

- 16.11 If the out-of-specification result has been obtained after release, the responsible clinician must be immediately informed to either prevent its administration or evaluate the possible consequences for patients.

## 17. Externally contracted quality control tests

- 17.1 Some quality control tests may be outsourced to external contractors where a facility does not have the capability or capacity to undertake these quality control tests. Contractors should be evaluated and qualified in accordance with a written procedure. Records of this evaluation process must be maintained.
- 17.2 A written quality agreement must be established, clearly outlining:
- the duties and responsibilities of each party;
  - the scope of the contracted work (for example, sterility testing);
  - any technical arrangements related to the contract;
  - audits of the contractor should be conducted as necessary ([1](#)).
- 17.3 The contracted organization is not permitted to subcontract any work entrusted to it without prior evaluation and approval by the laboratory.

## 18. Safety

### 18.1 Personnel safety

- 18.1 Safety in the quality control laboratory for radiopharmaceuticals is often underestimated, yet it should be a top priority. Although the radioactivity levels handled in these laboratories are relatively low, most manipulations are performed manually, increasing the risk of radiation exposure and contamination. Therefore, strict safety measures must be in place to minimize unnecessary worker exposure and contamination risks ([2](#)).
- 18.2 All personnel working in the quality control laboratory must wear personal dosimetry devices and appropriate personal protective equipment at all times ([5](#), [6](#)). Personnel leaving the quality control laboratory should undergo radioactive contamination monitoring, and measures should be established to prevent the spread of contamination from the laboratory.
- 18.3 Special precautions must be taken when handling alpha-emitting radionuclides, due to their high emission energy and short penetration range. While they pose a lower risk of external radiation exposure, they are difficult to detect, and ingestion or inhalation can result in serious

radiation hazards. Extra surface contamination control measures should be implemented to minimize the risk of ingestion or inhalation.

- 18.4 Special precautions, such as respirator use, may be required when testing samples of volatile or aerosol-forming test samples.
- 18.5 General and specific safety instructions, reflecting identified risk, should be made available to all staff members and regularly updated as needed. Training materials may include:
  - written guidelines;
  - poster displays;
  - audiovisual materials;
  - safety seminars and workshops ([1](#)).

## 18.2 Environmental safety

- 18.6 The handling of quality control samples should be conducted in a shielded fume hood or, at minimum, behind a suitable shielded protective barrier. The thickness of the shielding must be determined based on the type and activity of the radionuclide being handled. Additionally, quality control equipment should be shielded when necessary (for example, HPLC column and waste HPLC eluent bottle).
- 18.7 Radioactive waste generated during quality control activities must be stored in suitable shielded containers until radioactive decay has sufficiently reduced its activity to safe levels.
- 18.8 Standard chemical safety requirements applicable to laboratories handling flammable, corrosive, toxic chemicals and gases must also be observed in radiopharmaceutical quality control laboratories.

## 18.3 General rules for safe laboratory practices

- 18.9 All personnel must adhere to national regulations and standard operating procedures to ensure safe working conditions. These include the following.
  - Safety data sheets should be available to staff before conducting any testing.
  - All staff should be trained in the proper use of fire extinguishers and on the safe handling of glassware, corrosive reagents and solvents.
  - Personnel should wear laboratory coats or other protective clothing, including eye protection.

- All chemical containers must be clearly labelled with hazard warning signages, such as:
  - “Poison”;
  - “Flammable”;
  - “Radioactive” (where applicable).
- First-aid kits should be available, and staff should be trained in:
  - first-aid techniques;
  - emergency medical response;
  - use of antidotes for chemical exposures.
- The appropriate protective clothing should be available, including:
  - eye protection (safety goggles);
  - face masks and gloves.
- Depending on the operations performed and the associated risk assessment, safety showers may need to be installed and maintained in reasonably close proximity.
- Eyewash stations should be available in close proximity to the areas where the work is being performed.
- Poisonous or hazardous substances must be identified and clearly labelled. However, staff should be reminded not to assume that other chemicals and biological materials are inherently safe.

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